

* Describe states by $|n, l, m\rangle$

2D: (x, y) or (r, ϕ)

$$\langle \psi | \psi \rangle = \int_{-\infty}^{\infty} \psi^*(\vec{r}) \psi(\vec{r}) d\vec{r}^2$$
$$= \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \psi^*(x, y) \psi(x, y) = 1$$

* Prob. of finding a particle in a region R of $\mathbb{M}^2$: $R \subset \mathbb{M}^2$

$$P_R = \int_R |\psi|^2$$

More concretely, for a rectangular region:
 $a < x < b$ & $c < y < d$

$$P_{a < x < b, c < y < d} = \int_a^b dx \int_c^d dy |\psi(x, y)|^2$$

* Since: $\int_{-\infty}^a + \int_a^b + \int_b^{\infty} = \int_{-\infty}^{\infty}$
Then the above is breaking up the probability (i.e. breaking up 1).

* Calculating components of an arbitrary wavefunction ψ in the direction of a particular eigenfunction with eigenvalues ψ_n we require:

$$\langle \psi_n | \psi \rangle = \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \psi_n^*(x, y) \psi(x, y)$$

This yields:

$$|\psi\rangle = \sum_{n=0}^{\infty} \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy \psi_n^*(x, y) \psi(x, y) |\psi_n\rangle$$

$$|l, m_l\rangle: \quad l = 0, 1, 2, \dots$$

$$m_l = -l, -l+1, \dots, 0, \dots, l-1, l$$

$$\vec{L}^2 |l, m_l\rangle = l(l+1) |l, m_l\rangle$$

$$L_z |l, m_l\rangle = m_l |l, m_l\rangle$$

$$L^2 = L_x^2 + L_y^2 + L_z^2$$

* Possible to represent orbital angular momentum ops and eigenstates in the position representation

$$L_z = -i\hbar \left(\hat{x} \frac{\partial}{\partial y} - y \frac{\partial}{\partial x} \right)$$

$$|l, m_l\rangle \longrightarrow |\varphi, p\rangle$$

* In spherical coordinates: (r, θ)

$$L_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$L^2 = -\hbar^2 \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right)$$

Ham op d Schrödinger:

$$\hat{H} = \frac{\hat{p}^2}{2\mu} + V(r)$$

$$\hat{p}^2 = \hat{p} \cdot \hat{p} = -\hbar^2 \nabla^2$$

$$\hat{H} \psi(r) = E \psi(r) \Rightarrow \frac{\hat{p}^2}{2\mu} (\psi(r)) + \hat{V}(r) \psi = E \psi$$

Dynamics: $i\hbar \frac{\partial}{\partial t} \psi(r, t) = \hat{H} \psi(r, t)$

$$\psi(r, t) = \sum_n C_n e^{iE_n t / \hbar} \phi_n(r)$$

$$C_n = \langle \phi_n | \psi_0 \rangle = \int \phi_n^* \psi(r, t_0) dV$$

$$L_z = -i\hbar \frac{\partial}{\partial \phi}$$

$$\hat{H} = -\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial}{\partial r} \right) + \frac{1}{2\mu} \frac{1}{r^2} \hat{L}_z^2 + \frac{1}{2} \mu \omega^2 r^2$$